VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS305

Theory of Computation and Automata - Comprehensive Syllabus Lecture Note

Syllabus Focus: CS305 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Theory of Computation and Automata. Includes

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Finite Automata & Regular Languages

" Syllabus Scope: DFA, NFA, epsilon-NFA, subset construction algorithm, minimization of DFA Myhill-Nerode

Unit 2: Regular Language Properties & Pumping Lemma

" Syllabus Scope: Pumping lemma for regular languages, closure properties union, intersection, complement

CS305: Theory of Computation and Automata - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Context-Free Grammars & Pushdown Automata

" Derivation trees, ambiguity, Chomsky Normal Form CNF, Greibach Normal Form GNF, PDA transition mechanics

Unit 4: Context-Free Language Limits & Pumping Lemma

" Pumping lemma for CFLs, Ogden lemma, closure properties of CFLs, CYK parsing algorithm

Unit 5: Turing Machines & Undecidability

" Standard Turing Machine model, multi-tape TMs, Church-Turing thesis, Universal Turing Machine, Halting probl

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Construct a minimal DFA accepting the set of binary strings where the count of 0s is divisible by 3 an
2. Using the Pumping Lemma for Regular Languages, prove that L = {0^n 1^n | n >= 0} is not regular.
3. Prove that the Halting Problem for Turing Machines is undecidable using a diagonal argument.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John E. Hopcroft, Rajeev Motwani - Introduction to Automata Theory, Languages, and Computation Pearson
* Michael Sipser - Introduction to the Theory of Computation Cengage Learning 3rd Ed

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